

Equivariant Systems Theory and Observer Design for Autonomous Systems

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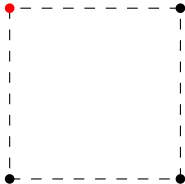
Outline

- 1 Symmetry
- 2 Lie groups
- 3 Lie group actions
- 4 Error variables

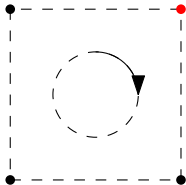
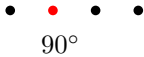
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Symmetry



Symmetry



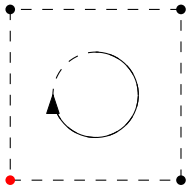
Symmetry



270°

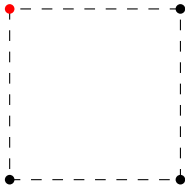
$90^\circ + 180^\circ$

-90°

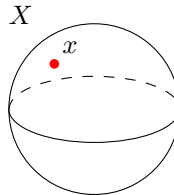
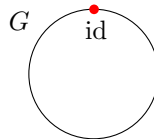


complete symmetry $\mathbb{Z}_4$

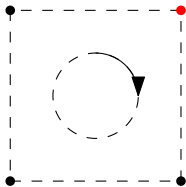
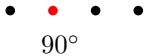
Symmetry



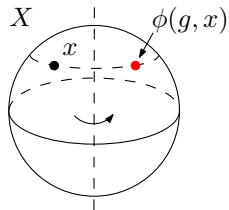
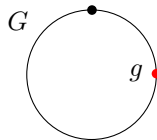
complete symmetry $\mathbb{Z}_4$



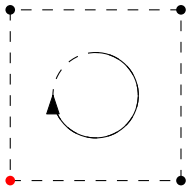
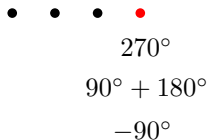
Symmetry



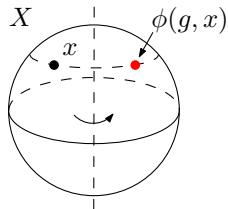
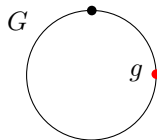
complete symmetry $\mathbb{Z}_4$



Symmetry



complete symmetry $\mathbb{Z}_4$



partial symmetry S^1
complete symmetry $SO(3)$

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Lie groups

A *Lie group* $\mathbf{G}$ is a differentiable (smooth) manifold that carries a group structure such that the group operations

$$\begin{aligned} \text{mult}: \mathbf{G} \times \mathbf{G} &\rightarrow \mathbf{G}, (X, Y) \mapsto X \cdot Y \\ &\text{and} \\ \text{inv}: \mathbf{G} &\rightarrow \mathbf{G}, X \mapsto X^{-1} \end{aligned}$$

are smooth maps.

Lie groups are models for *continuous symmetries*, e.g. the group $\text{SO}(3)$ of rotations, or the group $\text{SE}(3)$ of rigid body transformations of 3D Euclidean space.

A *matrix Lie group* is a closed subgroup of the general linear group $\text{GL}(n)$ of all real $n \times n$ invertible matrices.

Matrix Lie groups

The multiplication operation of a matrix Lie group is matrix multiplication, inversion is matrix inversion and the identity element is the identity matrix.

To check that $SO(3)$ is a (matrix) Lie group, we just need to check that the product of two rotation matrices is a rotation matrix, the inverse of a rotation matrix is a rotation matrix, and the limit of a sequence of rotation matrices is a rotation matrix.

$$SO(3) = \{X \in GL(3) \mid XX^T = I, \det(X) = 1\}$$

Realisations of $\mathbf{G}$

One and the same abstract Lie group $\mathbf{G}$ can be *realised* in different algebraic ways. These group realisations are all *isomorphic* as Lie groups.

$$\mathrm{SE}(3) = \left\{ \begin{pmatrix} R & p \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4 \times 4} \mid R^\top R = I, \det(R) = 1, p \in \mathbb{R}^3 \right\}$$

$$\mathrm{SE}(3) = \{(R, p) \mid R^\top R = I, \det(R) = 1, p \in \mathbb{R}^3\}$$

$$\mathrm{SE}(3) = \left\{ \begin{pmatrix} R & p & 0 \\ 0 & 1 & 0 \\ 0 & 0 & R \end{pmatrix} \in \mathbb{R}^{7 \times 7} \mid R^\top R = I, \det(R) = 1, p \in \mathbb{R}^3 \right\}$$

Realisations of $\mathbf{G}$

When we realise $\mathbf{G}$ as a matrix Lie group, we do not need to explicitly provide the formulas for group multiplication and inversion, they are matrix multiplication and matrix inversion. There are Lie groups that do not have a matrix realisation, for example the *Heisenberg group*.

For the previous realisation of $\text{SE}(3)$:

$$(R_1, p_1) \cdot (R_2, p_2) = (R_1 R_2, R_1 p_2 + p_1), \quad (R, p)^{-1} = (R^\top, -R^\top p)$$

Product groups

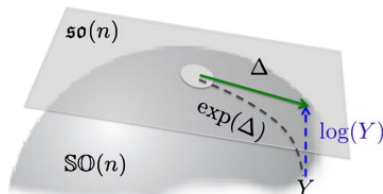
$SE(3)$ is an example of a *semi-direct product* of two Lie groups, namely $SE(3) = SO(3) \ltimes \mathbb{R}^3$.

The *direct product* $SO(3) \times \mathbb{R}^3$ is a different (non-isomorphic) Lie group:

$$(R_1, p_1) \cdot (R_2, p_2) = (R_1 R_2, p_1 + p_2), \quad (R, p)^{-1} = (R^T, -p)$$

We can always form new Lie groups by taking products of existing Lie groups.

The Lie algebra of $\mathbf{G}$



The tangent space $T_I \mathbf{G}$ at the identity element of a Lie group $\mathbf{G}$ naturally carries the structure of a *Lie algebra* $\mathfrak{g}$, a vector space with an additional operation, the *Lie bracket* $[\cdot, \cdot]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$.

For a matrix Lie group the Lie bracket is the *matrix commutator* $[X, Y] = XY - YX$.

For $\mathbf{G} = \text{SO}(3)$, the Lie algebra is the set $\mathfrak{so}(3)$ of all skew-symmetric real 3×3 matrices.

Left and right translation

The group multiplication defines two important diffeomorphisms L_X and R_X per element $X \in \mathbf{G}$:

$$L_X: \mathbf{G} \rightarrow \mathbf{G}, Y \mapsto X \cdot Y$$

and

$$R_X: \mathbf{G} \rightarrow \mathbf{G}, Y \mapsto Y \cdot X$$

The differentials of left and right translation at the identity naturally map the Lie algebra isomorphically onto the tangent space at X . This is directly visible in the formulas for matrix Lie groups.

$$dL_X(I)U = XU \text{ and } dR_X(I)U = UX$$

The adjoint operator

For $X \in \mathbf{G}$ the *adjoint operator* is the Lie algebra automorphism

$$\text{Ad}_X: \mathfrak{g} \rightarrow \mathfrak{g}, \quad U \mapsto dR_{X^{-1}}(X) \circ dL_X(I)U$$

For a matrix Lie group, $\text{Ad}_X(U) = XUX^{-1}$.

The adjoint operator is the differential of the inner group automorphism

$$L_X \circ R_{X^{-1}} = R_{X^{-1}} \circ L_X: \mathbf{G} \rightarrow \mathbf{G}$$

It gives rise to the *adjoint representation* $\text{Ad}(\mathbf{G}) \subset \text{End}(\mathfrak{g})$ of $\mathbf{G}$ on the vector space $\mathfrak{g}$.

Invariant vector fields

The vector field $X \mapsto XU$, $X \in \mathbf{G}$ for a fixed $U \in \mathfrak{g}$ is obviously left invariant (unchanged by left translation). Equally, the vector field $X \mapsto UX$ is right invariant.

We have $XU = XUX^{-1}X = \text{Ad}_X(U)X = VX$.

A parameterized family of these vector fields gives rise to an invariant control system

$$\dot{X} = XU \text{ resp. } \dot{X} = VX$$

For $\mathbf{G} = \text{SO}(3)$ these are the kinematic equations for a rigid body with angular velocity U (resp. V) in $\mathfrak{so}(3)$.

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Right group actions

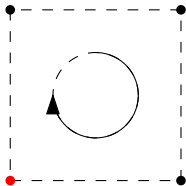
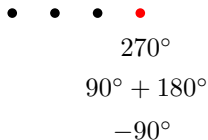
A *right group action* ϕ of the Lie group $\mathbf{G}$ on a smooth manifold $\mathcal{X}$ is a smooth map

$$\phi: \mathbf{G} \times \mathcal{X} \rightarrow \mathcal{X}$$

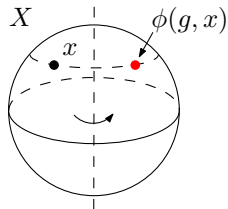
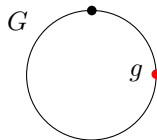
such that $\phi(I, x) = x$ and $\phi(A, \phi(B, x)) = \phi(BA, x)$ for all $A, B \in \mathbf{G}$ and $x \in \mathcal{X}$.

Intuitively, a group action moves points around the manifold, and an object that is defined on the manifold and does not change after application of the action is “symmetric”, with the symmetry described by the structure of the Lie group and the action map.

Symmetry



complete symmetry $\mathbb{Z}_4$



partial symmetry S^1
complete symmetry $SO(3)$

Transitivity

A group action ϕ gives rise to two families of smooth maps

$$\phi_A: \mathcal{X} \rightarrow \mathcal{X}, \quad \phi_A(x) = \phi(A, x)$$

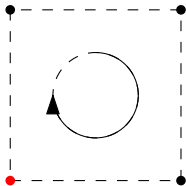
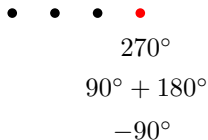
and

$$\phi_x: \mathbf{G} \rightarrow \mathcal{X}, \quad \phi_x(A) = \phi(A, x)$$

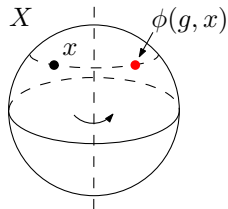
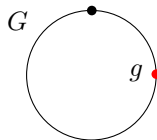
Note that ϕ_A is always a diffeomorphism.

The group action is called *transitive* if ϕ_x is surjective and in this case $\mathcal{X}$ is called a *homogeneous space* of $\mathbf{G}$.

Symmetry



complete symmetry $\mathbb{Z}_4$



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Homogeneous spaces

The *stabilizer* of an element $x \in \mathcal{X}$ is the subgroup of $\mathbf{G}$

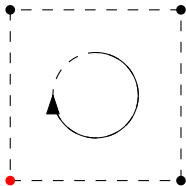
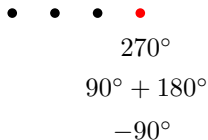
$$\text{stab}_\phi(x) = \{A \in \mathbf{G} \mid \phi(A, x) = \phi_x(A) = x\}$$

For a homogeneous space $\mathcal{X}$ all stabilizers are isomorphic as subgroups of $\mathbf{G}$ and the submersion $\phi_x: \mathbf{G} \rightarrow \mathcal{X}$ induces a diffeomorphism

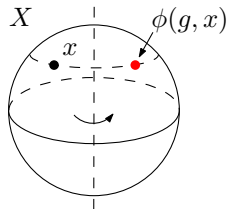
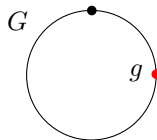
$$G / \text{stab}_\phi(x) \rightarrow \mathcal{X}$$

The 2-sphere $S^2 \subset \mathbb{R}^3$ is a homogeneous space of $SO(3)$ with right action $\phi(R, x) = R^\top x$.

Symmetry



complete symmetry $\mathbb{Z}_4$



partial symmetry S^1
complete symmetry $SO(3)$

Invariant control systems revisited

A Lie group $\mathbf{G}$ acts on itself by right translation

$$\phi: \mathbf{G} \times \mathcal{G} \rightarrow \mathcal{G}, \quad \phi(A, X) = XA$$

creating a homogeneous space $\mathcal{G}$.

This action separates the group structure (symmetry) out from the differential structure of $\mathbf{G}$, hence it is common practice to think of $\mathcal{G}$ as a $\mathbf{G}$ -*torsor* rather than a Lie group. We indicate that with the change of notation from $\mathbf{G}$ to $\mathcal{G}$.

It is now an easy exercise to show that the control system $\dot{X} = XU = (XA)(A^{-1}U)$ is “symmetric” with respect to two right actions of $\mathbf{G}$ on $\mathcal{G}$ resp. $\mathfrak{g}$.

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Canonical group errors

As a means to compare trajectories on a $\mathbf{G}$ -torsor $\mathcal{G}$ we introduce the *canonical type I error*

$$E_I: \mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G}, \quad E_I(\hat{X}, X) = \hat{X}X^{-1}$$

Observe that the partial maps $E_I(\cdot, X): \mathcal{G} \rightarrow \mathcal{G}$ and $E_I(\hat{X}, \cdot): \mathcal{G} \rightarrow \mathcal{G}$ are diffeomorphisms.

The canonical type I error is right invariant (“symmetric”).

$$E_I(R_S \hat{X}, R_S X) = E_I(\hat{X}S, XS) = \hat{X}SS^{-1}X^{-1} = E_I(\hat{X}, X)$$

Later, we will also introduce the left invariant canonical type II error E_{II} .

Output errors

Augment the system $\dot{X} = XU$ on the $\mathbf{G}$ -torsor $\mathcal{G}$ with an *output*

$$y = \rho(X, \mathring{y})$$

where $\rho: \mathbf{G} \times \mathcal{Y} \rightarrow \mathcal{Y}$ is a group action and $\mathring{y}$ is a reference measurement.

We can then define the *type I output error*

$$e^I: \mathcal{G} \times \mathcal{Y} \rightarrow \mathcal{Y}, \quad e^I(\hat{X}, y) = \rho(\hat{X}^{-1}, y).$$

This error has a natural equivariance (“symmetry”)

$$e^I(R_S \hat{X}, \rho(S, y)) = e^I(\hat{X}, y).$$

This type of error will be crucial for the design and analysis of observers that are implementable from sensor measurements.